Long-range epigenetic concordance — analysis workflow

single-cell NOMe-HiC · GCH = GpC accessibility · HCG = CpG methylation · Δ_{signed} = (co-open − co-closed): long-range contact − short-range baseline

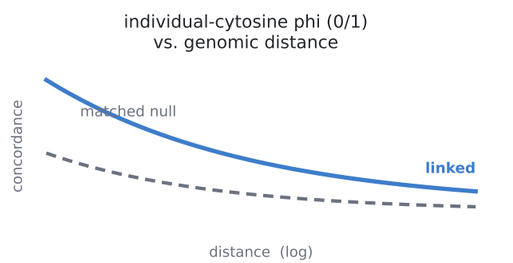
INPUT



single-cell NOMe-HiC BAMs · reference genome (.2bit) · dbSNP (optional)

MEASURE

site_distance_correlation

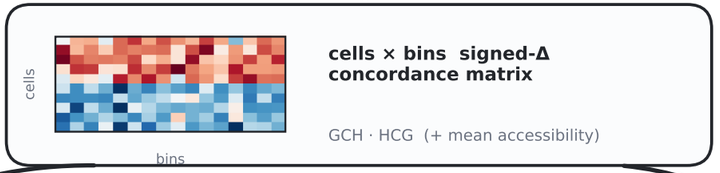


GCH·GCH HCG·HCG HCG·GCH
Δ = linked − null, bootstrap

nome_hic_concordance

run per context: GCH + HCG

1. GCH / HCG contexts (excl. GCG, dbSNP)
2. parse HQ distal pairs (mapQ>30, ≥20 kb)
3. β-binomial mixture (EM)
4. per-end posterior P(open|k,n)
5. signed Δ per (cell, bin)
6. empirical-Bayes shrinkage
7. norm · RWR impute · SVD → UMAP



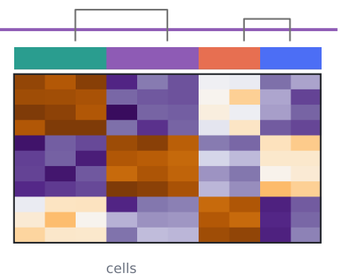
cells × bins signed-Δ concordance matrix

GCH · HCG (+ mean accessibility)

INTERPRET

celltype_marker_bins

bins with high between-cell-type variance, low within-type batch variance → marker heatmap



control: non-selected bins still separate types (perm. p)

nome_hic_compare

regress Δ on 1D mean accessibility + 3D scHi-C compartments

$\Delta = \beta_{1D}\,\text{access} + \beta_{3D}\,\text{comp} + \textbf{residual}$

does the residual still cluster?



residual UMAP → not reducible to 1D or 3D alone